Skills Summary

One-Variable Inequalities: Reversing the Symbol

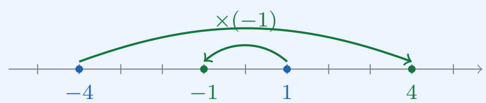
Use this reference while completing your worksheet or when studying.

1. Multiply or divide by a negative (multiply-or-divide-by-negative)

The rule: multiplying or dividing *both sides* by a **negative** number **reverses** the symbol. Adding or subtracting anything — including a negative — does *not*.

$>$ becomes $<$ $<$ becomes $>$ $\geq$ becomes $\leq$ $\leq$ becomes $\geq$

Why: multiplying by a negative *reflects* the number line through 0. Whatever was on the left lands on the right, so "smaller" and "larger" trade places.



Before: $-4 < 1$. After the reflection: $4 > -1$. Same two numbers, opposite order.

Example: Solve $-6x < 18$

Step 1. The variable is multiplied by -6, so undoing it means dividing by a negative — that is the one move that reverses the symbol.

Step 2. $\dfrac{-6x}{-6} \overset{\text{flip}}{>} \dfrac{18}{-6} \quad \Rightarrow \quad \boldsymbol{x > -3}$

Step 3. Check with a number from the set: $x = 0$ gives $-6(0) = 0 < 18$ ✓.

Try it: Solve $-2x \geq 10$

check: $\boldsymbol{x \leq -5}$

2. Isolate first, then flip (isolate-then-flip)

The order: (1) add or subtract to get the x-term alone — this step *never* flips; (2) divide by the coefficient — this step flips *only if* that coefficient is negative.

A negative sign inside the problem is not enough. Ask the narrower question: **"am I dividing by a negative right now?"**

Example: Solve $5 - 4x \leq 17$

Step 1. Two moves are needed and only the second one can flip, so subtract first and decide about the symbol later.

Step 2. Subtract 5: $-4x \leq 12$ (symbol unchanged — this was a shift).

Step 3. Divide by -4: $\dfrac{-4x}{-4} \overset{\text{flip}}{\geq} \dfrac{12}{-4} \quad \Rightarrow \quad \boldsymbol{x \geq -3}$

Try it: Solve $2 - 5x > 22$

check: $\boldsymbol{x < -4}$

3. Diagnose a missing flip (diagnose-and-justify-the-flip)

The two-second test: pick any number from the answer's solution set and put it into the **original** inequality.

- If it makes a true statement, the answer is probably right.
- If it makes a *false* statement, and the arithmetic checks out, the symbol is backwards — a missing flip.

Test a number *far* from the boundary (-10, not -2.9): a wrong symbol shows up loudest out there.

Example: Ana solved $-8x > 24$ and wrote $x > -3$. Diagnose and fix.
Step 1. Her arithmetic ($24 \div -8 = -3$) is right, so test her solution set rather than re-doing the division.
Step 2. Take $x = 0$ from her set: $-8(0) = 0$, and $0 > 24$ is false — her set holds non-solutions, so the symbol is backwards.
Step 3. Dividing by -8 reverses it: $\boldsymbol{x < -3}$

Try it: Rob solved $-\dfrac{x}{2} \leq 4$ and wrote $x \leq -8$. Fix it.

check: $\boldsymbol{x \geq -8}$

(!) Watch out: A negative sign somewhere in the problem does not by itself flip anything. In $x - 7 > -2$ you add 7 to both sides and the symbol stays: $x > 5$. The flip is caused only by *multiplying or dividing* both sides by a negative.